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1 Closure of Integers under Addition, Subtraction, and Multiplication

Let $a, b \in \mathbb{Z}$. We prove that:

- $a + b \in \mathbb{Z}$
- $a - b \in \mathbb{Z}$
- $ab \in \mathbb{Z}$

1.1 Representation of integers

Every integer is either:

- 0,
- a natural number, or
- the negative of a natural number.

Thus, we can write integers in the form: $a = \pm m, b = \pm n$, where $m, n \in \mathbb{N} \setminus \{0\}$.

1.2 Closure under addition

We consider cases.

1.2.1 Case 1: $a, b \geq 0$

Then $a, b \in \mathbb{N}$. Since $\mathbb{N}$ is closed under addition,

$$a + b \in \mathbb{N}.$$

1.2.2 Case 2: $a = -m, b = -n$

$$a + b = (-m) + (-n) = -(m + n).$$

Since $m + n \in \mathbb{N}$, it follows that $a + b \in \mathbb{Z}$.

1.2.3 Case 3: $a = m, b = -n$

$$a + b = m - n.$$

If $m \geq n$, then $m - n \in \mathbb{N} \subseteq \mathbb{Z}$. If $m < n$, then:

$$m - n = -(n - m),$$

and $n - m \in \mathbb{N}$.

Hence, $a + b \in \mathbb{Z}$.

1.3 Closure under subtraction

Subtraction reduces to addition:

$$a - b = a + (-b).$$

Since $b \in \mathbb{Z}$, we have $-b \in \mathbb{Z}$. From closure under addition,

$$a - b \in \mathbb{Z}.$$

1.4 Closure under multiplication

Again, consider cases.

1.4.1 Case 1: $a, b \geq 0$

Then $ab \in \mathbb{N} \subseteq \mathbb{Z}$.

1.4.2 Case 2: $a = -m, b = n$

$$ab = (-m)n = -(mn).$$

Since $mn \in \mathbb{N}$, we have $ab \in \mathbb{Z}$.

1.4.3 Case 3: $a = -m$, $b = -n$

$$ab = (-m)(-n) = mn.$$

Thus $ab \in \mathbb{Z}$.

1.5 Conclusion

For all $a, b \in \mathbb{Z}$:

$$a + b \in \mathbb{Z}, a - b \in \mathbb{Z}, ab \in \mathbb{Z}.$$

Hence, the integers are closed under addition, subtraction, and multiplication.